L_j = Latest start time of event j

t_{ij} = Duration of activity (i,j)

The next step after making the time estimates is the calculations which are done in the following ways:

(i) Forward Pass Calculations

(ii) Backward Pass Calculations.

Forward pass calculations:

We start from the initial node 1 with starting time of the project as zero. Proceed through the network visiting nodes in an increasing order of node number and end at the final node of the network. At each node, we calculate earliest start times for each activity by considering E_i as the earliest occurrence of node i.

The method may be summarized as below:

Step 1: Set $E_1 = 0$; $i = 1$ (initial node).

Step 2: Set the earliest start time(EST) for each activity that begins at node i as $ES_{ij} = E_i$; for all activities (i, j) that start at node i.

Step 3: Compute the earliest finish time (EFT) of each activity that begins at node i by adding the earliest start time of the activity to the duration of the activity. Thus $EF_{ij} = ES_{ij} + t_{ij} = E_i + t_{ij}$.

Step 4: Move on to next node, say node $j(j > i)$ and compute the earliest start time at node j, using $E_j = \max i\{EF_{ij}\} = \max_i \{E_i + t_{ij}\}$ for all immediate predecessor activities.

Step 5: If $j = n$ (final node), then the earliest finish time for the project is given by $E_n = \max\{EF_{ij}\} = \max\{E_{n-1} + t_{ij}\}$.

Backward pass calculations:

We start from the final (last) node n of the network, proceed through the network visiting nodes in the decreasing order of node numbers and end at the initial node 1. At each node, we calculate the latest finish time and latest start time for each activity by considering L_j as the latest occurrence of node j. The method may be summarized below:

Step 1: $L_n = E_n$; for $j = n$

Step 2: Set the latest finish time (LFT)of each activity that ends at node j as $LF_{ij} = L_j$

Step 3: Compute the latest start time (LST) of all activities ending at node j, subtracting the duration of each activity from the latest finish time of the activity. Thus, $LS_{ij} = LF_{ij} - t_{ij} = L_j - t_{ij}$

Step 4: Proceed backward to the next node i($i<j$) in the sequence and compute the latest occurrence time at node i using $L_i = \min_j \{LS_{ij}\} = \min_j \{L_j - t_{ij}\}$.

Step 5: If $j = 1$ (initial node), then $L_1 = \min\{LS_{ij}\} = \min\{L_2 - t_{ij}\}$

Critical path:

The longest path connected by the activities in the network is called the critical path. A path along which it takes the longest duration.

For the activity (i,j), to lie on the critical path, following conditions must be satisfied:

(i) $E_i = L_i$ and $E_j = L_j$

(ii) $E_j - E_i = L_j - L_i = t_{ij}$

Example 10.13

Compute the earliest start time, earliest finish time ,latest start time and latest finish time of each activity of the project given below:

10_11th_BM-STAT_Ch-10-EM.indd 247 21-04-2020 12:32:55 PM